

Executive Summary

This abstract outlines the research scope and expected outcomes for the Lane Keep Assist (LKA) sub-system, developed as a contribution to the groups Advanced Driver Assistance systems (ADAS) project. LKA is a critical safety feature designed to prevent unintentional lane departures by applying corrective steering when a vehicle drifts outside its lane. However, current LKA systems which rely on camera based vision, experience significant performance degradation due to inadequate lane markings and adverse weather conditions. To address this issue, this project explores a Vehicle-to-infrastructure (V2I) solution approach using Chip Enabled Raised Pavement Markers (CERPMs), which transmit positional data to the vehicle directly.

The project focuses on evaluating the effectiveness of different interpolation algorithms for estimating the lane center, while also analysing the impact various CERPM dropout scenarios would have on system accuracy. A simulation based methodology will be implemented using Python, incorporating real world road geometries. The outcomes of this research includes a comparative analysis of interpolation algorithms and an assessment of the interpolation systems robustness under infrastructure failure.

Table Of Contents

Executive Summary	1
Statement of Problem.....	1
Introduction.....	1
Background.....	2
Relevance and importance	2
Background.....	2
Literature review	2
Strategy	3
Project Management	3
Gnatt Chart.....	3
Deliverables	4
References.....	Error! Bookmark not defined.

Statement of Problem

Lane Keep Assist is considered one of the most common and fundamental components of ADAS. LKA and hence ADAS relies heavily on accurate detection of lane boundaries to operate reliably and safely.

Introduction

LKA is a driver assistance feature that applies corrective steering to prevent unintentional crossing of lane boundaries (Lane departure). Unlike automated centering systems, LKA only activates reactively when a vehicle begins to drift towards a lane edge without a turn signal active. It is intended that the driver is easily able to override the steering correction and maintain full control over the vehicle.

Current commercial LKA systems use front facing cameras with computer vision algorithms to detect lane boundaries, but these systems tend to degrade substantially on roads with poor lane markings or adverse weather conditions.

Recent research has proposed a few methods to reduce this degradation. Most interesting however is a proposed and proved V2I solution could be used alone or in conjunction with current vision algorithms to eliminate the degradation all together. The researchers use Chip Enabled Raised Pavement Markers to transmit GPS coordinates directly to the onboard computer where interpolation is used on the left and right CERPMs to calculate the lanes boundaries. However, research was not conducted into different interpolation methods nor to the effect that sequential missing CERPMs would have on the estimations of boundaries.

This research aims to address this gap by conducting research into the effects that different interpolation algorithms as well as how various types of CERPM dropout will affect the calculation of the lanes center line.

Background

Relevance and importance

In Australia, Lane departure remains one of the most significant contributors to road fatalities. A fact sheet from the government states that 62% of fatalities nationally were due to lane departure (Run off road and head on). 73% in regional areas, 71% in remote areas and 41% in major cities [3]. Similarly, in the United States (US), the National Highway Traffic Safety Administration (NHTSA) estimated that on average a LKA equipped car was 24% less likely to be involved in a fatal road departure crash. [4]. These statistics demonstrate that LKA is key system in reducing fatalities in road crashes.

The European Commission mandated that all new motor vehicles sold after the 7th of July 2024 were required to have LKA in addition to various other safety features [5]. Additionally, LKA forms a foundational component of ADAS upon which various higher level autonomous driving function are built.

Background

While lane departure warning (LDW) systems alert drivers to unintended lane departures, they rely on the driver taking reacting and taking action in a timely manner; Something that can be unreliable due to driver fatigue, distraction and/or impairment. LKA systems address this problem by actively applying steering to prevent the vehicle from crossing lane lines. Despite wide commercial adoption and deployment, real world LKA performance remains inconsistent. Field research shows that LKA systems performance reduces substantially under poor lane markings, poor weather and near lane transitions such as merges and diverges [6].

LKA technology has advanced considerably over the past decade. Early implementations used simple threshold-based lane detection with rule-based torque overlays. Current systems use deep learning base lane detecting combined with Kalman filtering for robust lane tracking. Control strategies have also advanced from basic torque overlay approaches to systems that share control authority adaptively between the driver and the system [9].

Literature review

Recent research has advanced both the perception and control aspects of LKA systems significantly. OpenLKA the first large scale, open, real world dataset was released. It includes just about 400 hours of LKA steered data from 62 production vehicle models collected through road testing in Florida and global contributors. Their empirical benchmark revealed that LKA systems degrade substantially under poor lane markings and near lane transitions [6].

A paper [7] and its follow up [8] proposed a Multiple Model Adaptive Estimation (MMEA) algorithm combining front and rear cameras with a Kalman filter to improve lane detection under challenging conditions such as night and adverse weather. camera to calculate the uncertainty of each model, and hence selected the most accurate model for the current conditions.

Deliverables

- **Analysis Program:** A complete program that extracts road geometry, simulates CERPM placement, applies configurable dropout, and evaluates multiple interpolation algorithms against the true road centreline.
- **Interpolation Algorithm Comparison:** Performance data for each algorithm across multiple different road geometries.
- **CERPM Dropout Analysis:** An analysis of how various types of CERPM dropout affects accurately defining the center of the lane.

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